Week 03 — Before You Come

100L · Semester 1 · AI in Software & Mobile Development

You have probably already used an AI chat assistant, such as ChatGPT, Gemini, Claude, or Meta AI inside WhatsApp. This week asks three honest questions about them: how they produce an answer, why they are sometimes **confidently wrong**, and what they do to your learning if you let them think for you. You will also meet the rules this programme follows about AI, and the reasons behind them.

This page takes about 45 minutes. Do the three tasks in order.

Task 1: The predictive-text game (5 min, no data needed)

Your phone's keyboard already contains a very small version of the idea behind AI chat assistants.

1. Open any app where you can type: Notes, or a message to yourself. **Don't send anything.**
2. Type `I am going to` and stop.
3. Look at the suggested words above the keyboard. Tap the **middle** suggestion.
4. Keep tapping the middle suggestion until you have about 15 more words.
5. Copy the sentence into your notebook, exactly as it came out.

Then answer, in one line each:

- Does the sentence sound like English?
- Is it *true*? Does your phone know where you are going?

No suggestions above your keyboard? Check that predictive text is switched on in your keyboard's settings, or try a friend's phone. If you can't do it, do Task 2 and read about it in class.

Task 2: My AI so far (10 min, no data needed)

Answer these in your notebook, a few lines each. Be honest; nobody marks this page.

1. Have you used an AI chat assistant? Which one, and for what? "Never" is a fine answer.
2. Describe one time an AI gave you an answer that was **wrong**, or that you weren't sure about. How did you find out? If it never happened, how would you know if it had?
3. If an AI can already write code, why do you think this programme still teaches you to write it yourself?
4. If **you** made the rules, should 100L students be allowed to use AI for their lab work? Why or why not?

Keep your answers. In class you will compare them with the programme's actual rules.

Task 3: Read and watch (about 30 min)

Do these in order. Each has a question to keep in mind. You don't need to write the answers down; the session uses them.

1. **3Blue1Brown: Large Language Models explained briefly** (video, about 8 min)
   https://www.youtube.com/watch?v=LPZh9BOjkQs

   This uses data. Set the video quality to the lowest option. Skip the parts about how training works inside the computer if they lose you; the first half is the part that matters.

   Keep in mind: each time the model adds a word, what is it actually choosing?
2. **MIT Sloan: When AI Gets It Wrong: Addressing AI Hallucinations and Bias** (10 min)
   https://mitsloanedtech.mit.edu/ai/basics/addressing-ai-hallucinations-and-bias/

   Read **"Inaccurate Content"** and **"Why is AI Flawed?"**. Skim the rest. Skip any terms you don't know, such as RAG or temperature.

   Keep in mind: is an AI chat tool built to tell the **truth**, or to sound **right**?
3. **Anthropic: How AI assistance impacts the formation of coding skills** (10 min)
   https://www.anthropic.com/research/AI-assistance-coding-skills

   Anthropic makes an AI coding assistant, and this is their own experiment with 52 programmers. Read down to the end of the results. You don't need to understand the statistics or the code library they used.

   Keep in mind: which group learned more, and on which kind of quiz question was the difference biggest?
4. **Stack Overflow Developer Survey 2025: AI** (5 min) https://survey.stackoverflow.co/2025/ai#2-accuracy-of-ai-tools

   Look only at the charts **"Accuracy of AI tools"** and, just below it, **"AI tool frustrations"**.

   Keep in mind: what share of developers say they **highly trust** what AI tools tell them?

Optional, if you want more:

- CS50 at Harvard: Academic honesty: how a famous beginner course sets its AI rules. Search the page for "AI".
- Simon Willison: Here's how I use LLMs to help me write code: an experienced developer's honest advice. Skip the code. Read **"Set reasonable expectations"**, **"Account for training cut-off dates"** and **"LLMs amplify existing expertise"**.
- Elements of AI: a free online course from the University of Helsinki. Chapter 1 explains what "AI" means without any maths.

No data this week? Tasks 1 and 2 matter more, and they need no connection. The session covers the readings' main points.

Bring with you

- Your **predictive-text sentence** and your **Task 2 answers**
- A notebook and a pen

- A phone, if you have one. You **don't** need an AI app or an account. If you already use one, bring it.
- A laptop if you have one, but it is **not** required. Tools are installed in week 4.

Under 18? AI services set a minimum age, and some need a parent's permission. You can do the whole session by working with your partner or watching the tutor's screen.